

BARKLEY, KY, USA

WMO: 724350

Lat: 37.056N

Lon: 88.774W

Elev: 413

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.9	(c) 15.9	(d) 0.2	(e) 5.6	(f) 13.6	(g) 5.2	(h) 7.3	(i) 18.8	(j) 23.2	(k) 41.8	(l) 20.3	(m) 42.4	(n) 6.7	(o) 20	(p) 0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.9	(c) 94.1	(d) 76.3	(e) 92.0	(f) 76.3	(g) 90.1	(h) 75.6	(i) 79.7	(j) 90.3	(k) 78.4	(l) 88.5	(m) 77.4	(n) 86.9	(o) 7.3	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.6	140.7	85.7	75.5	135.5	84.6	74.4	130.6	83.3	43.3	90.2	42.2	88.8	41.0	86.8	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.2	17.1	15.4	DB	3.8	97.9	6.7	3.0	-1.0	100.1	-4.9	101.9	-8.7	103.6	-13.5	105.8	
			WB	2.9	81.7	6.5	1.2	-1.8	82.6	-5.6	83.3	-9.3	84.0	-14.0	84.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	35.5	39.7	48.5	58.8	68.0	76.1	78.8	77.7	70.8	59.7	47.8	39.2
	DBStd	17.62	11.84	11.31	10.54	8.91	7.56	5.36	4.66	4.72	6.92	8.83	9.67	10.52
	HDD50	1513	474	322	158	25	0	0	0	0	16	155	363	
	HDD65	3995	916	710	520	223	61	2	0	1	26	214	521	801
	CDD50	4604	23	33	110	290	557	784	893	859	624	316	88	27
	CDD65	1609	0	1	7	37	152	337	428	395	200	49	3	0
	CDH74	15772	0	1	46	299	1300	3352	4418	3946	1931	462	17	0
	CDH80	6362	0	0	2	37	380	1377	1952	1748	752	113	1	0
(14) Wind	WSAvg	6.5	8.0	8.2	8.3	8.0	6.5	5.5	4.9	4.4	4.6	5.8	6.8	7.3
(15) Precipitation	PrecAvg	48.6	3.3	3.8	4.5	4.9	5.0	4.2	4.2	3.0	3.8	3.3	4.2	4.4
	PrecMax	65.5	10.0	13.3	14.9	14.5	9.3	11.0	12.5	7.2	9.2	7.4	10.5	11.5
	PrecMin	36.3	0.6	0.9	0.9	1.5	1.4	0.4	0.3	0.4	0.1	0.0	0.9	0.6
	PrecStd	7.7	2.1	2.4	2.6	2.8	2.3	2.6	2.8	1.7	2.2	1.8	2.4	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.3	71.5	79.1	83.7	89.8	94.7	97.5	98.0	94.1	87.4	76.3	69.4
		MCWB	61.3	62.2	63.9	66.9	73.0	75.6	78.3	75.8	72.5	71.2	63.9	63.0
	2%	DB	64.0	67.0	74.3	80.7	87.0	92.0	93.8	94.1	91.0	83.2	71.7	64.4
		MCWB	59.0	58.8	61.5	65.8	71.5	75.5	78.0	76.0	73.8	68.0	61.2	58.9
	5%	DB	59.5	62.6	70.3	77.6	84.4	90.1	91.6	91.3	88.0	79.4	68.0	60.2
		MCWB	55.0	55.2	59.0	64.0	70.1	75.1	77.8	76.1	72.2	65.7	59.3	55.1
	10%	DB	54.3	58.1	66.0	74.3	81.6	87.8	89.5	88.8	84.5	75.2	64.0	56.0
		MCWB	49.0	51.5	56.4	62.4	69.2	74.1	76.4	75.2	70.0	63.7	56.5	51.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.6	63.8	66.2	70.9	76.2	79.9	81.3	81.1	77.3	73.8	67.0	63.7
		MCDB	66.0	69.5	75.4	79.6	85.6	90.1	91.7	91.5	88.4	84.4	72.7	68.3
	2%	WB	59.9	60.3	63.5	68.6	74.0	78.3	80.0	79.4	75.5	71.0	64.1	60.0
		MCDB	63.2	65.1	72.0	77.2	83.2	88.6	90.6	89.8	86.1	79.1	69.7	63.9
	5%	WB	56.0	57.0	60.8	66.3	72.3	77.0	78.8	78.0	74.1	68.6	60.9	56.1
		MCDB	59.0	61.5	68.4	74.4	81.5	86.8	89.1	87.8	83.9	75.8	66.7	59.4
	10%	WB	50.0	51.9	57.9	64.2	70.7	75.6	77.7	76.8	72.6	66.0	57.4	51.4
		MCDB	53.5	57.2	64.9	71.9	79.2	84.6	87.2	85.7	81.2	73.1	62.3	55.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.6	19.7	21.1	19.8	19.5	18.9	20.6	22.9	22.4	19.5	16.1
		MCDBR	19.2	22.7	23.8	23.3	21.8	21.5	20.7	23.2	25.5	25.7	22.8	19.7
	5% WB	MCWBR	16.0	17.0	14.6	12.3	9.9	8.5	8.2	9.1	10.2	12.6	15.1	16.0
		MCWBR	17.4	19.9	21.0	19.4	18.7	18.8	19.4	21.0	20.7	19.5	17.5	15.9
(40) Clear-Sky Solar Irradiance	taub		0.318	0.328	0.365	0.400	0.428	0.440	0.478	0.476	0.410	0.360	0.324	0.314
	taud		2.460	2.413	2.370	2.274	2.251	2.228	2.147	2.140	2.323	2.444	2.490	2.477
	Ebn at Noon		276	287	285	279	272	268	257	254	266	272	272	270
	Edh at Noon		27	31	36	42	43	44	48	47	37	30	26	24
(44) All-Sky Solar Radiation		RadAvg	684	876	1217	1598	1802	2025	1955	1793	1503	1138	780	582
		RadStd	65	109	106	101	153	128	120	93	92	125	83	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+173
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page